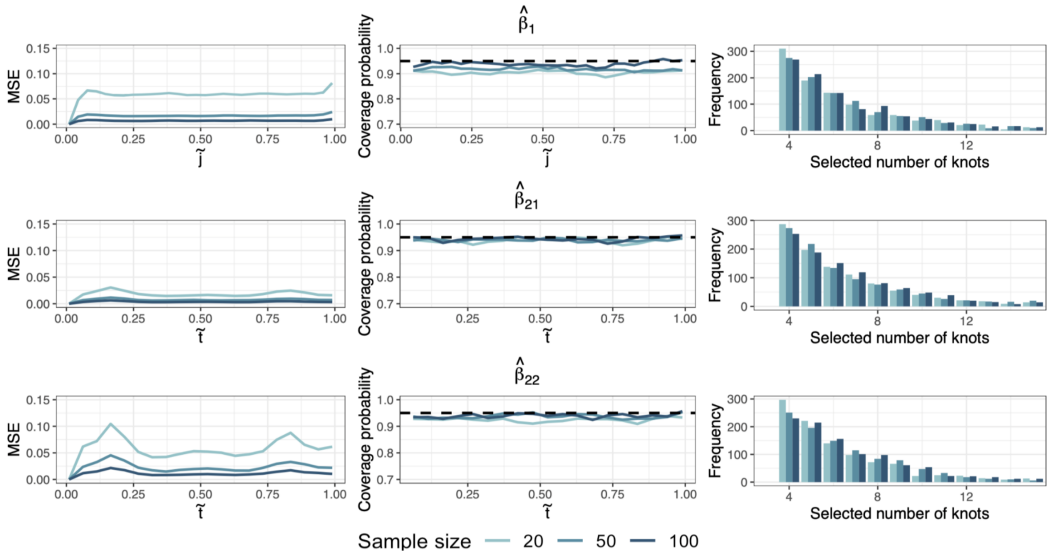

Figure 4: Mean squared error (left), coverage probability of 95% confidence interval (center), and selected number of knots (right) for $\hat{\beta}_1(\tilde{j})$, $\hat{\beta}_{21}(\tilde{t})$, and $\hat{\beta}_{22}(\tilde{t})$ under the simple function setting in the simulation.

7 Application

We assessed the effect of activity suggestions in the HeartSteps MRT on participants' minute-level step count using the proposed method. 37 participants were included in the analysis with 7629 decision points. Even though seven participants had less than 210 decision points of data (with reasons detailed in Klasnja et al. (2018)), the same number of knots and knot locations were used for everyone.

Because an activity suggestion may either be a walking suggestion or an anti-sedentary suggestion, we used a generalized version of the proposed model (detailed in Section D of the Supplementary Material to accommodate categorical treatments. Let A_{ij} take three values: 0 if no push notification was delivered (with probability 0.4), 1 if an activity suggestion was delivered (with probability 0.3), and 2 if an anti-sedentary suggestion was delivered (with probability 0.3). Let the proximal outcome Y_{ijt} be the log-transformed step count during

the t-th minute following decision point j of participant i. Let I_{ij} denote participant i's availability at decision point j: participants were considered unavailable at a decision point, and thus no treatment would be delivered, if they were currently driving, had manually turned off the intervention, were exercising 90 seconds before the decision point, or did not have an active internet connection (Seewald et al., 2019). Among the 7629 decision points, 6137 were available.

The original primary analysis of HeartSteps focused on a 30-minute aggregate step count proximal outcome (Klasnja et al., 2018). To assess how soon an individual responds to the suggestion, we consider the 60-minute time window following each decision point, i.e., $T = 60$. This also helps to assess whether the 30-minute window was sufficiently wide to capture the effect of the activity suggestions in the original analysis.

We first assessed how the CEE of the walking suggestions and the anti-sedentary suggestions varied within the 60-minute window and throughout the study. Then we investigated how the CEE is moderated by three time-varying variables: location, weekday/weekend, and recent activity level. Those were among the variables used to tailor the content of the suggestions (Smith et al., 2017).

Lastly, we mimic what a typical MRT analysis would be by examining aggregate outcomes, which further highlights the added value of examining the more granular longitudinal functional outcome.

7.1 Marginal Causal Excursion Effect

We considered the following CEE model for the walking suggestion ($A_{ij} = 1$) and the anti-sedentary suggestion ($A_{ij} = 2$), compared to no suggestion ($A_{ij} = 0$):

$$\mathbb{E}\{Y_{ijt}(\bar{A}_{i,j-1}, m) - Y_{ijt}(\bar{A}_{i,j-1}, 0) \mid I_{ij} = 1\} = \beta_0^{(m)} + \beta_1^{(m)}(j) + \beta_2^{(m)}(t), \quad m = 1, 2. \tag{9}$$

Here, $\beta_0^{(m)}$ represents the effect of the corresponding suggestion ($m = 1$ for walking, $m = 2$ for anti-sedentary), compared to no suggestion, on the log-transformed step count at the first minute following the first decision point (recall the identifiability constraints $\beta_1^{(m)}(1) = \beta_2^{(m)}(1) = 0$). $\beta_1^{(m)}(j)$ captures how the effect of the corresponding suggestion varies over decision points. $\beta_2^{(m)}(t)$ captures how the effect of the corresponding suggestion varies over 60 minutes following a decision point.

The Stage 1 nuisance function $\eta(j, t, H_{ij})$ was fitted using a generalized additive model with penalized splines, implemented by the `gam` function in the `mgcv` R package (Wood, 2017). We included as features the minute index t, the decision point index j, location (an indicator for being at home/work or not), the weekday indicator, recent activity level (an indicator of whether an individual took less than 56 steps in the 30-minute window prior to a decision point), and individual's log-transformed cumulative step counts in a 30-minute window following the previous decision point. The Stage 2 functional parameters $\beta_1^{(m)}(j)$ and $\beta_2^{(m)}(t)$ were modeled using cubic spline. The number of knots was selected through a one-pass coordinate search using 10-fold cross-validation, and the selected number of interior knots for $\hat{\beta}_1^{(1)}(j), \hat{\beta}_2^{(1)}(t)$, $\hat{\beta}_1^{(2)}(j)$, and $\hat{\beta}_2^{(2)}(t)$ were 3, 0, 0, and 5, respectively.

Figure 5 shows the trajectories of CEE over decision points and minutes for the walking suggestion and the anti-sedentary suggestion. For both types of suggestions, peaks of the effects (the yellow region in Figure 5(a) and (b)) occurred at the beginning of the study and